

Welcome!



**Have you ever felt
like your training
wasn't producing results?**



If yes! Meet Apex



Your AI-powered personal trainer



Apex, maximizing your training



- **Aaron Ricca**
 - Electrical and computer engineering
 - Ex professional skier
 - Climber
- **Derya Özsoy**
 - Informatik
 - Kickboxing



What's the problem ?

- training is far from optimized
- hard to find good exercises
- time is always too little



How does **Apex** work?

- the End User enters his **data**
- Apex learns with the collected **User data** and **data** from Internet and tries to find common patterns.
- At the end the User receives a personalized training plan



Data time !

Input attributes:

- Objective
- Available time
- Favorite exercises
- Biological attributes (Height, body fat%, medical condition, etc.)
- Alimentation (Cuisine, Diet type, enhancing substances, etc.)

Output attributes:

- Training periodization
- List of exercises
- Alimentation improvement tips
- Corresponding improvement %
[Continuous]



Data Sources

Research & Literature

- Scientific papers
- Studies & publications
- Medical research

Real-World Data

- Gyms & fitness centers
- Professional coaches
- Testers & athletes

User-Generated

- Surveys & questionnaires
- Smartwatch data
- Fitness app tracking

Continuous Growth

- Data collected from users



Data Preprocessing

- Convert different units & scales to SI units
 - Height (feet → cm), Weight (lb → kg), etc.
- **Standardization** → for classification algorithms (KNN, Random Forest)
- **Normalization** → for regression algorithms



Few data?

→ More data?



Synthetic data!



SMOTE

Synthetic Minority Oversampling Technique

- Used for **imbalanced datasets**
→ e.g., 90% fit people and 10% overweight
- Problem: ML models **bias the majority class**
- SMOTE **creates new synthetic data** of the minority class



How SMOTE Works

Our scenario: 10% overweight, 90% fit

-**User A:** Age 35, Weight 95kg, Beginner level

-**User B:** Age 40, Weight 100kg, Beginner level

SMOTE finds two **similar overweight users** and creates a new realistic profile:

-**Synthetic user C:** Age 37, Weight 97kg, Beginner level ✨

✅ Makes the minority class **denser and more balanced** (27%)



What methods are there and what are we using for Apex?

Unsupervised Learning

✗ No labeled data needed

- looks for inherent similarities and differences within the data on its own

Semi-supervised Learning

- between supervised and unsupervised Learning
- has sizable amount of unlabeled data with a small amount of labeled

Supervised Learning

✓ labeled data needed

- learns patterns and relationships between input and output data.
- learns to recognize patterns in order to make predictions for new, unknown and similar data.



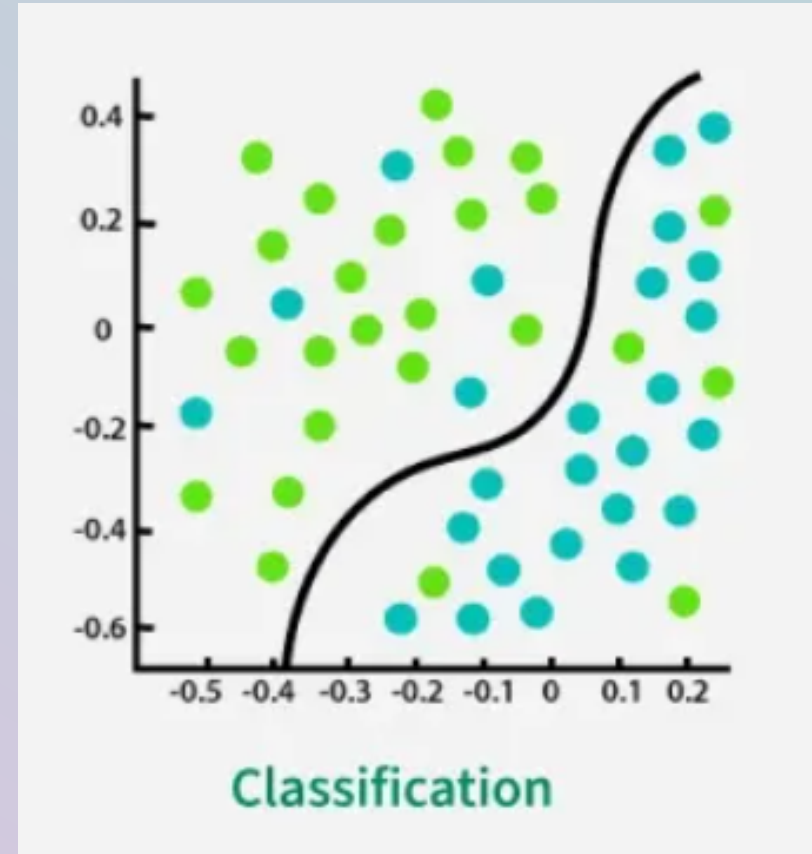
Tools for Supervised Learning

Classification

- predict an outcome or event in the future.
- outcome can take two possible values such as True or False, Default or no Default, yes or no.

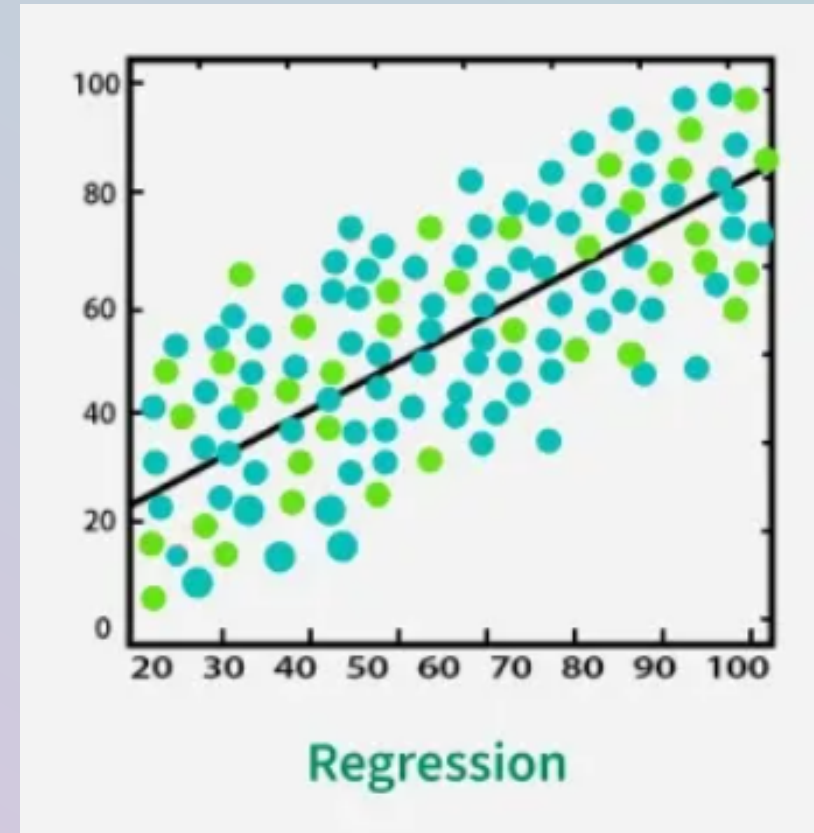
→ exp: We categorise some Data "*sick*" an "*healthy*" so

that the user receives a training plan that does not



Regression

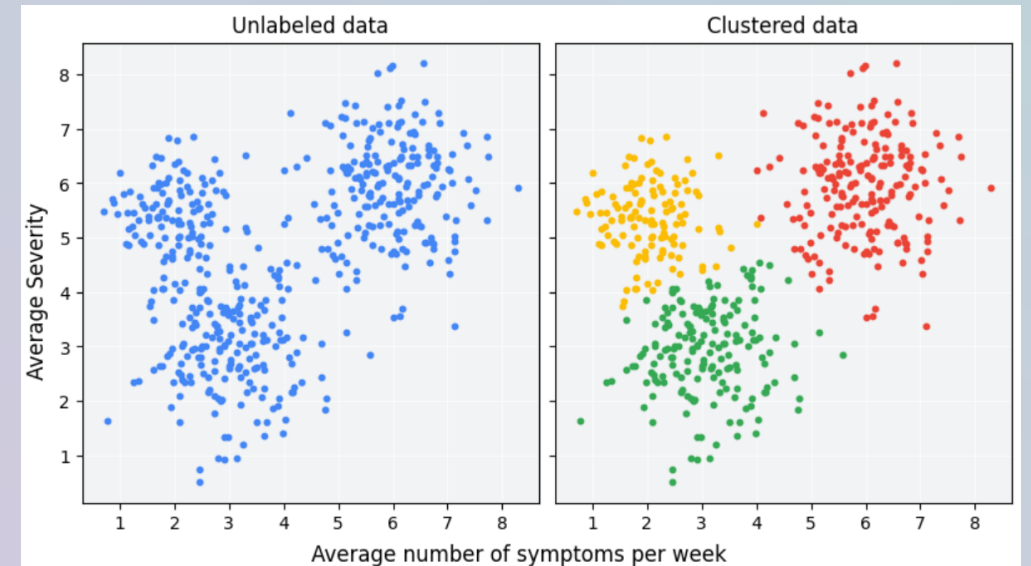
- predict continuous values such as sales, salary, weight, or temperature.
→ exp: predict a good Trainingsplan for the User



Unsupervised Learning Methods

Clustering

used to identify patterns, segment data, and simplify complex datasets by grouping similar items together.



Association rules

used to discover interesting relationships and dependencies between items in large datasets.

The rules are expressed as "if-then" statements, such as $X \rightarrow Y$



Dimensionality reduction

Reduces high dimensions to fewer features.

Commonly used algorithms are PCA (Principal Component Analysis) and SVD (Singular Value Decomposition)

Retains important information while removing unnecessary complexity. It's useful for Data visualization (2D/3D) and Model simplification.



Helps avoid the “curse of dimensionality” (too many features

Supervised Learning Algorithms

Linear Regression

- Prediction of continuous values (e.g., weight based on height).
- Uses a linear relationship between input and target variables.
- Corresponds to the "line of best fit."

Logistic Regression

- The target variable is binary (e.g., yes/no, 1/0).
- Uses the logit function to predict probabilities.
- Ideal for classification problems.



Decision Tree

- Makes decisions using a tree-like structure.
- Each leaf represents a decision or prediction.
- Easy to interpret and visually understandable.

K Nearest Neighbor

- Categorizes data based on proximity to other points.
- The new point is assigned to the class of its nearest neighbors.
- Simple, but computationally intensive for large data sets.



Random Forest

- Combination of many decision trees (ensemble method).
- Provides more accurate and stable results.
- Can be used for classification and regression.

Neural Networks

- Inspired by the human brain.
- Consist of layers of neurons (input, hidden layers, output).
- Learn by adjusting weights based on errors (backpropagation).
- Can recognize complex patterns and nonlinear relationships.

Naive Bayes

- Based on Bayes' theorem (probabilities).

- Assumption: Features are



Data Processing

- **SMOTE** - Balance underrepresented groups
- **Cross-Validation** - Ensure model generalization
- **PCA (Principal component analysis)** - Reduce the data dimensionality



Why is Apex better?

- **Open source** - Transparent algorithms, community improvements
- **Data-driven optimization** - Personalized vs generic programs
- **Lower injury risk** - Biomechanical constraints and progressive overload
- **First of its kind** - Novel ML-based personal training approach
- **Continuous learning** - Improves with more user data



Possible risks, barriers and obstacles

- **Incomplete data can lead to incorrect results.**
 - can lead to injuries due to incorrect training suggestions
- **Time and resources.**
 - Data must always be updated and maintained.
 - Possible data loss.
 - High costs und delays.
- **Data breach.**



Ethical Considerations

- **Privacy & GDPR compliance** - Sensitive biometric and health data
- **Labor market impact** - Potential displacement of personal trainers
- **Economic effects** - Reduced gym memberships
- **Algorithmic bias** - Training data skewed toward young, fit persons
 - Risk: Inappropriate recommendations for underrepresented groups
 - Can lead to injuries in elderly or different body types
- **Dual-use concerns** - Technology repurposed beyond original intent
 - Example: Military training



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Apex: Any questions?

We're happy to dive deeper.

